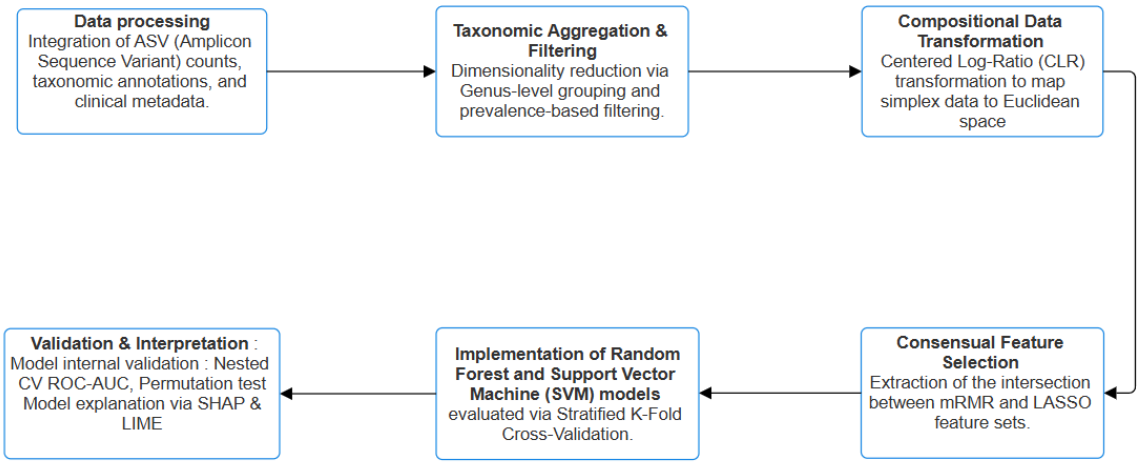
An ML approach to 16S rRNA microbiome biomarkers for predicting colorectal cancer

I. Abstract & Project objectives

Colorectal Cancer (CRC) is a leading cause of cancer-related mortality globally. While early detection at the precancerous adenoma stage significantly improves survival rates, traditional screening methods such as colonoscopy are invasive, and fecal immunochemical tests (FIT) exhibit limited sensitivity for early-stage lesions.
Recent literature highlights the critical role of gut microbiome dysbiosis in the pathogenesis of CRC. Consequently, this project develops a comprehensive bioinformatics and machine learning pipeline to identify non-invasive microbial biomarkers from stool samples. The specific objectives are:

- To systematically address the inherent challenges of microbiome amplicon sequencing data : specifically high dimensionality, sparsity —using robust mathematical transformations prior to modeling.
- To identify a minimal, discriminative subset of bacterial genera that differentiates between CRC patients, individuals with adenomatous polyps, and healthy controls without functional redundancy or model overfitting.
- To utilize eXplainable AI (XAI) frameworks, including SHAP and LIME, to elucidate the biological directionality of the selected biomarkers, transitioning the predictive model from a "black box" to a clinically transparent diagnostic tool.

II. Pipeline architecture overview



III. Detailed methodology and implementation

3.1. Data processing and taxonomic alignment

The primary data source for this study is the "CRC Gut Microbiome ML Data" dataset, publicly available via Kaggle. The dataset is derived from 16S rRNA gene sequencing and was rigorously preprocessed using the DADA2 pipeline to resolve exact biological sequences rather than traditional clustered OTUs. The repository is structured into three interdependent files:

- **seqtab_nochim_export.xlsx (ASV Abundance Matrix):** This file contains the quantitative read counts, structured with 60 patient samples as rows and distinct Amplicon Sequence Variants (ASV DNA sequences) as columns.
- **taxa_species_export.xlsx (Taxonomic Dictionary):** A mapping table that aligns each unannotated ASV DNA sequence from the abundance matrix to its corresponding taxonomic lineage across seven evolutionary ranks (Kingdom, Phylum, Class, Order, Family, Genus, Species).
- **metadata.csv (Clinical Metadata):** This file provides the clinical context for each sample, including patient identifiers (SampleID), demographic covariates (Sex, Age), and the primary categorical target variable (host_disease), which stratifies the 60 samples into three distinct classes: 'Colorectal cancer', 'Adenomatous Polyps', and 'Healthy'.

To construct the primary feature matrix:

- The ASV count matrix was transposed (.T) to align ASV sequences as the index and Sample IDs as columns.
- An inner join (pd.merge) was executed between the transposed count matrix and the taxonomy table using the ASV DNA sequences as the primary key.
- To preserve biological relevance and mitigate extreme sparsity, the sequence-level data was aggregated to the taxonomic Genus level using the pandas .groupby('Genus').sum() function. Unclassified taxa were imputed using their respective Family-level nomenclature appended with the suffix _unclassified.
- The aggregated matrix was transposed back to the standard machine learning format (Samples x Features). A critical data alignment step involved casting the SampleID indices to strings and stripping whitespaces to successfully execute an inner join with the metadata.csv file, resulting in a cohesive df_ml_ready dataframe comprising 59 valid samples and 247 bacterial genera.

3.2. Compositional data transformation (CLR)

Microbiome sequencing yields compositional data; read counts are constrained by the arbitrary sequencing depth of the instrument, meaning the data resides in a constrained simplex space.
Standard statistical methods assume unconstrained Euclidean space, and applying them directly to relative abundances induces spurious negative correlations.
To resolve this, the pipeline implemented the Centered Log-Ratio (CLR) transformation:

- Taxa present in fewer than 10% of the samples were removed. This thresholding eliminated transient environmental contaminants and technical noise, reducing the feature space from 247 to 110 core genera.
- A pseudocount of 1 was added uniformly to the filtered matrix to satisfy the logarithmic constraints of the CLR function (to avoid taking the log(0) in the CLR formula)
- The skbio.stats.composition.clr function was applied. For a sample vector x containing D genera, the CLR transforms the abundance of genus j as follows: where g(x) is the geometric mean of all features in the sample. This operation successfully mapped the bounded counts into continuous, symmetric real numbers (in the result table: ranging from -2.55 to 8.65), satisfying the assumptions of downstream linear and distance-based algorithms.

3.3. mRMR and LASSO Double-filter

Given the ratio of features (110) to samples (59), direct classification would result in

overfitting. A dual-algorithm feature selection strategy was employed:

- **mRMR (Minimum Redundancy Maximum Relevance):** This algorithm evaluates features sequentially. It maximizes the mutual information between the bacterial genus and the clinical target (host_disease), while simultaneously minimizing the average mutual information (redundancy) with previously selected genera. This ensures that the selected taxa represent distinct biological pathways rather than co-abundant clusters.
- **LASSO (Least Absolute Shrinkage and Selection Operator):** A Logistic Regression model was fitted using an L1 penalty term (penalty='l1', solver='liblinear'). The L1 regularization mathematically forces the coefficients of non-predictive or highly collinear features to exactly zero, effectively performing embedded feature selection.

mRMR was configured to select the top 15 features, while LASSO identified 49 features with non-zero coefficients. The intersection of these two sets yielded **15 Consensual Biomarkers**. The intersection guarantees that the retained biomarkers are both highly predictive and strictly independent.

3.4 Model initialization & Hyperparameter selection

Two distinct classification algorithms—Random Forest (RF) and Support Vector Machine (SVM)—were instantiated to handle the multiclass target vector (host_disease: CRC, Adenoma, Healthy). The hyperparameters were strictly constrained to penalize complexity:

Random Forest Classifier (RandomForestClassifier):

- n_estimators=100: Establishes a sufficiently large ensemble of decision trees to ensure statistical stability without incurring unnecessary computational overhead.
- max_depth=4: By strictly limiting the depth of the trees, the model is forcibly prevented from capturing sample-specific noise (memorization). It is constrained to learn only the most prominent, generalized decision boundaries.
- min_samples_split=5: Ensures that a node must contain at least 5 samples to undergo a further split, acting as a secondary mechanism against terminal-node overfitting.
- class_weight='balanced': Automatically adjusts the algorithmic weights inversely proportional to class frequencies in the input data.

Support Vector Machine (SVC):

- kernel='linear': Because the prior CLR transformation successfully projected the simplex compositional data into an unbounded Euclidean space, the biological features are appropriately scaled for linear separability. A linear kernel is highly resistant to overfitting in this transformed space compared to polynomial or RBF kernels.
- class_weight='balanced': Applies identical class-frequency penalization as utilized in the Random Forest.

3.5. Model internal validation

Standard cross-validation is insufficient if feature selection is applied to the entire dataset prior to splitting, as it induces Data Leakage (the feature selector "sees" the test data). To enforce strict methodological rigor, a nested validation architecture was implemented:

- **StratifiedKFold:** The data was partitioned into 5 folds (n_splits=5, shuffle=True). Given n=59, each fold contains approximately 11-12 samples. Stratification ensures that the original underlying prevalence of CRC, Adenoma, and Healthy patients is exactly mirrored in every training and validation subset, preventing skewed evaluation metrics.
- **Nested CV AUC-ROC** : To eradicate data leakage, the LASSO feature selection and the Random Forest classifier were encapsulated within a unified Scikit-Learn pipeline. During the 5-Fold CV, the LASSO algorithm mathematically re-evaluates and selects features based *exclusively* on the training data of that specific fold. The test fold

remains completely blind to the model until the final prediction step. The saga solver was explicitly chosen to support robust L1 regularization convergence across 2000 iterations.

- To assess the reproducibility of the biological signals, the pipeline tracked the exact feature subset selected by LASSO across all 5 folds. Biomarkers that are selected in 5/5 folds represent "Highly Stable Biomarkers"—indicating a persistent physiological relationship independent of data partitioning.
- **Permutation Testing (permutation_test_score):** To validate the statistical significance of the classifier, a permutation test was executed with 1000 iterations. The null hypothesis (H0) posits that the microbiome features and the clinical labels are independent. By randomly shuffling the target labels 1000 times and recalculating the nested CV ROC-AUC, an empirical null distribution was generated.

IV. Results & Model interpretation

Baseline vs. Nested cross-validation performance

```
 Random Forest Performance (5-Fold CV)
Accuracy : 0.709 ± 0.178
ROC-AUC  : 0.828 ± 0.121
F1-Macro : 0.712 ± 0.173

 Support Vector Machine Performance (5-Fold CV)
Accuracy : 0.677 ± 0.171
ROC-AUC  : 0.840 ± 0.103
F1-Macro : 0.690 ± 0.160
```

Accuracy, ROC-AUC, and F1-Macro from the initial cross_validate run

```
Nested CV ROC-AUC (OVR): 0.802 ± 0.142

Highly stable biomarkers selected in 5/5 folds:
 - Akkermansia
 - Christensenellaceae R-7 group
 - Fusicatenibacter
 - Incertae Sedis
 - Lachnospira
 - Lachnospiraceae UCG-010
 - Lachnospiraceae_unclassified
 - NK4A214 group
 - Peptostreptococcus
 - Ruminococcus
```

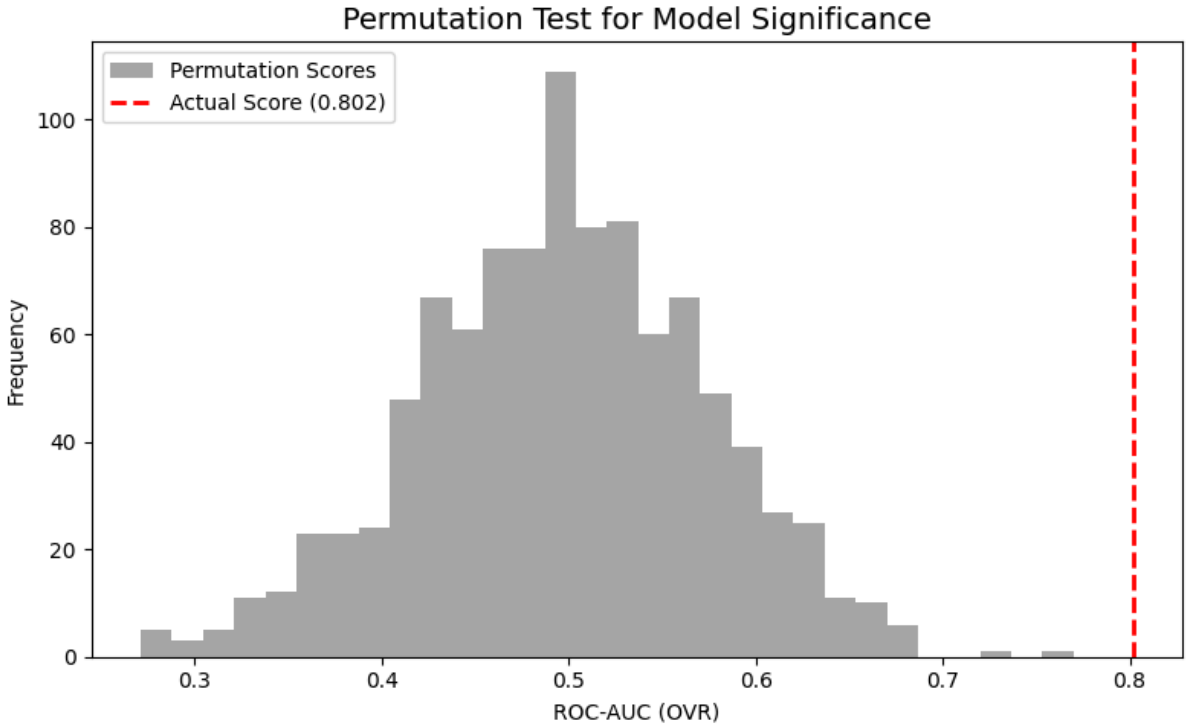
Nested CV ROC-AUC (OVR) and Highly stable biomarkers across folds

In the initial baseline 5-Fold Cross-Validation, feature selection was performed prior to data splitting, resulting in over-optimistic ROC-AUC scores of 0.828 (Random Forest) and 0.840 (SVM).

The unbiased Nested CV yielded a highly robust ROC-AUC of 0.802 ± 0.142. This slight, expected adjustment validates that the model's predictive power is driven by genuine biological signals rather than algorithmic overfitting or memorization of the test sets.

Permutation Testing

To definitively confirm that the model's predictive performance was not a statistical artifact or a result of overfitting on a small cohort (n=59), an empirical permutation test was applied using 1,000 randomized iterations.



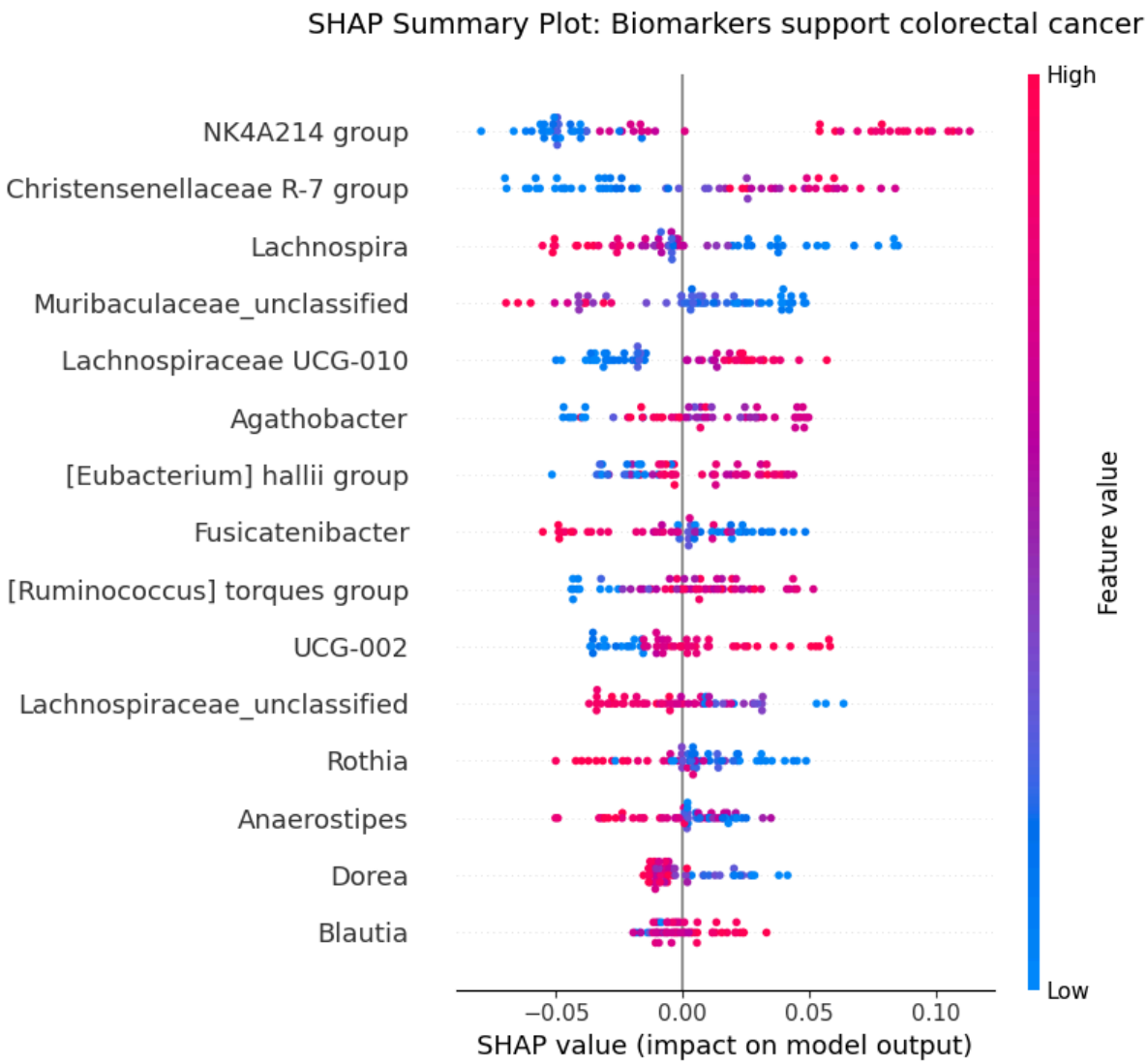
Permutation n=1000 result

The model's actual performance (red dashed line at ROC-AUC = 0.802) is situated far to the right, completely disjointed from the randomized null distribution (gray bars)

This provides mathematical proof that the 15 consensual microbial biomarkers identified by the pipeline contain genuine signals for Colorectal Cancer and Adenoma, and the model's predictive power is driven by true biological relationships rather than random noise.

Biological Interpretability via SHAP and LIME

The integration of SHAP uncovers the directional pathophysiology learned by the Random Forest



SHAP output

Taxa at the top of the chart, such as the *NK4A214 group*, *Christensenellaceae R-7 group*,.., act as strong risk factors. As seen in the plot, high abundances of these bacteria (the red dots) consistently fall on the right side of the zero line (positive SHAP values). This means their presence directly drives the model to predict CRC.

On the flip side, genera like *Lachnospira*, *Agathobacter*, *Fusicatenibacter*, and *Anaerostipes* show a clear protective effect. High levels of these bacteria (red dots) group heavily on the left side of the chart (negative SHAP values), actively lowering the predicted risk of CRC.

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patient explain: CRC1
label: Colorectal cancer
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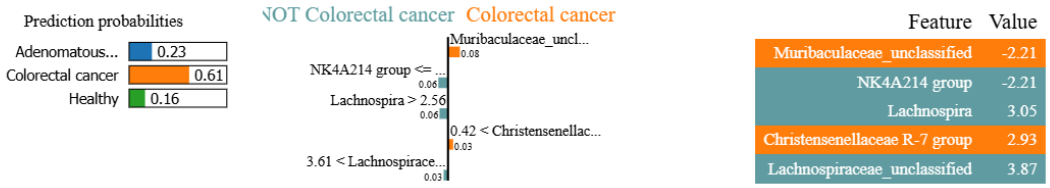

LIME output

While SHAP provides a global overview of the model's logic, Local Interpretable Model-agnostic Explanations (LIME) isolates the decision pathway for individual instances.

For the selected patient CRC1 (patient_idx = 'CRC1'), the Random Forest predicted a 61% probability of Colorectal Cancer. The LIME tabular explainer distinctly outlines the contributing thresholds: the prediction was heavily driven by an abnormal depletion of *NK4A214 group* and *Muribaculaceae_unclassified* (pushing the probability up), while simultaneously being mitigated by elevated levels of *Lachnospira* (pushing the probability down).

V. Current limitations

Clinical double-check for the SHAP output :

All the microbial taxa: *Ruminococcaceae NK4A214 group*, *Christensenellaceae R-7 group, Lachnospira*, *Agathobacter*, *Fusicatenibacter*, and *Anaerostipes* are generally considered beneficial commensal bacteria that *inhibit* the development of CRC

=> Model vs. Biological Reality conflict.

Potential causes :

- The entire dataset consists of only 59 patients (roughly 20 per group), with this small sample size, ML models are highly susceptible to learning random noise. In this specific group of 59 people, the CRC patients might have coincidentally had higher levels of *NK4A214* and *Christensenellaceae R-7* due to unrecorded confounding factors—such as their specific hospital diet, their age, regional geography, or antibiotic usage prior to the study. The Random Forest model blindly recognized this mathematical pattern and used it to predict CRC, completely unaware that it violates global human biology.
- Because gut bacteria levels are so deeply linked, data models can easily misassign blame. If a good bacteria (*NK4A214*) surges to help fight off a real pathogen (*Fusobacterium*) in a colon cancer patient, the two variables become mathematically entangled. The algorithm (mRMR) might filter out the actual pathogen to reduce redundancy, leaving the model to point the finger at the helpful bacteria just because it was found at the scene of the crime.

Limitations regarding multi-cohort External Validation

Applying the trained model to external metagenomic datasets (e.g., from public repositories) frequently results in feature misalignment. The primary training dataset relies on the SILVA taxonomic database, which retains specific alphanumeric designations for uncultured clades. Conversely, external cohorts frequently utilize NCBI or mOTU reference databases. This discrepancy causes feature extraction algorithms to impute zeroes for unrecognized taxa, artificially degrading external predictive performance.

To transition this pipeline toward a universally deployable diagnostic tool, future architectures must integrate a Taxonomic Harmonization module. By establishing a translation matrix that maps SILVA-specific phylogenies to standard NCBI equivalents prior to feature selection.